

Modeling Complex Semantics Relations with Contrastively Fine-Tuned Relational Encoders

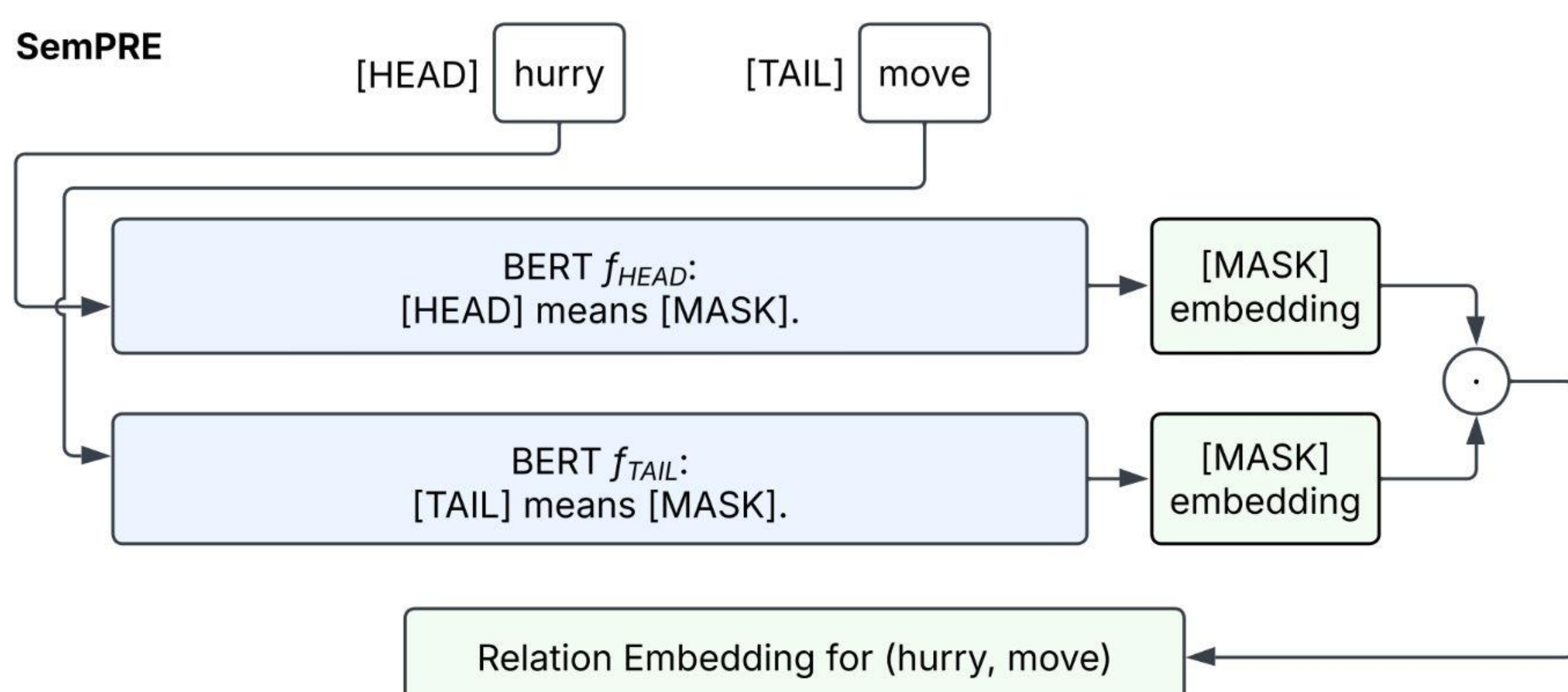
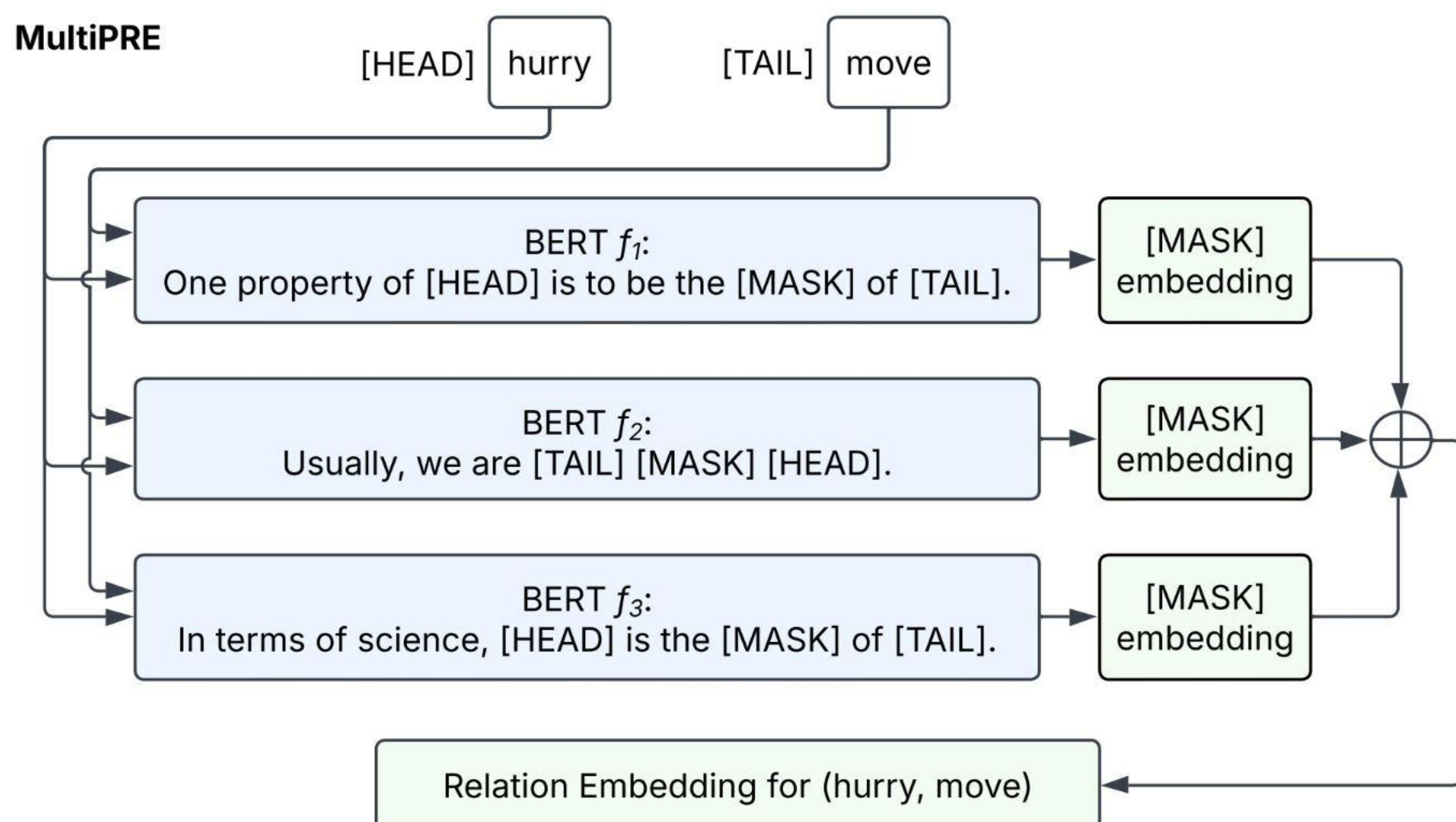
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01 INTRODUCTION

Understanding how words relate to each other like Paris is the capital of France, or hot is the opposite of cold is essential for many NLP tasks. These lexical relations support applications like: Relation extraction, Knowledge graph completion, Question answering, Story generation. However, existing methods have limitations: Language models tend to focus on a single relation and ignore others. Knowledge graphs rely on predefined schemas, which restrict relational diversity. Our goal is to **build compact, rich, and generalizable representations of these semantic relations**.

03 ENCODERS

We propose three relational encoders: UniPRE, MultiPRE and SemPRE. UniPRE uses a single prompt to extract a relation embedding from masked language modeling. MultiPRE combines multiple prompts to capture richer relational patterns. SemPRE encodes the semantics of head and tail separately, focusing on their individual meanings.



05 CONCLUSION

In this work, we introduced a new approach to encoding semantic relations using contrastive learning and diverse prompt templates. Our models, especially MultiPRE, outperform existing baselines such as RelBERT on both analogy and classification tasks. We also presented SemRelLM, a new dataset of over 80,000 LLM-generated triplets covering a wide range of relation types, including commonsense and functional relations often missing in traditional resources.

02 SemRelLM

We leverage large language models as a source of common-sense knowledge to create a semantically diverse dataset for training relational encoders. We generate triplets (head, relation, tail) that capture 16 types of semantic relationships.

To assess the quality of the generated data, we conducted a human evaluation over a sample of 200 triplets. Annotators rated each example on grammaticality, factual correctness, and relational plausibility. The results confirmed that most triplets are well-formed, semantically coherent, and representative of fine-grained lexical relations.

SemRelLM contains over 80,000 triplets, covering a much broader and more diverse range of relations than traditional datasets. It provides a rich foundation for learning generalizable relation embeddings.

Relation category	Unique triplet tokens	Unique triplet lemmas	Unique relation tokens	Unique relation lemmas
Antonymy	5003	4861	44	44
Attribute	5015	4885	5	4
Commonsense	5019	3902	1303	527
Functional	5026	4604	732	661
Collocation	5029	4779	1	1
Troponymy	5014	4883	2	2
Causal	5017	4358	486	317
Entailment	5006	4178	1	1
Spatial	5000	4892	108	106
Temporal	5003	4762	155	133
Hypernymy	5019	4971	218	205
Hyponymy	5001	4935	114	98
Meronymy	5015	4347	17	5
Member-collection	5019	4986	12	10
Synonymy	5014	4961	626	620
Similarity	5039	4852	1	1
All	80239	74576	3021	1811

04 EVALUATION

Model	U2	U4	BATS	Google	SCAN	NELL	T-REX	CN	Average
RelBERT-large	67.0	63.0	80.0	95.0	27.0	65.0	63.0	47.0	63.37
RelKB(RelBERT finetuning data)									
UniPRE-large	61.84	60.65	79.66	92.8	19.74	77.17	68.85	47.9	63.58
MultiPRE-large	61.84	67.36	82.32	93.6	20.3	76.33	70.49	53.52	65.72
SemPRE-large	31.58	32.64	49.58	53.6	11.51	74.67	56.28	26.85	42.09
SemRelLM(LLM-generated data)									
UniPRE-large	73.68	75.0	79.1	92.0	20.36	56.17	69.95	37.92	63.02
MultiPRE-large	70.61	74.31	80.77	92.2	23.21	62.67	64.48	37.16	63.18
SemPRE-large	37.72	40.51	49.58	54.8	15.53	54.0	16.39	23.32	36.48
RelKB+SemRelLM									
UniPRE-large	72.81	74.54	82.66	93.6	20.3	73.83	71.04	40.69	66.18
MultiPRE-large	71.05	73.15	83.77	94.6	25.56	69.83	65.57	39.51	65.38
SemPRE-large	38.6	40.05	57.53	62.2	16.96	69.83	38.25	24.83	43.53

Accuracy (in %) on analogy questions. RelBERT results are from (Ushio et al., 2023).

Model	BLESS	CogALexV	EVALution	K&H+N	ROOT09	Average
RelBERT-large	81.25	72.14	53.45	88.59	75.67	74.22
RelKB (RelBERT finetuning data)						
UniPRE-large	88.14	78.86	67.24	93.48	83.67	82.28
MultiPRE-large	90.54	86.82	64.94	94.37	83	83.93
SemPRE-large	80.93	69.4	41.38	92.74	69.67	70.82
SemRelLM(LLM-generated data)						
UniPRE-large	88.46	78.86	64.94	93.41	89.33	83
MultiPRE-large	92.63	85.07	67.24	94.37	89.67	85.8
SemPRE-large	79.33	69.9	47.13	93.11	78.0	73.49
RelKB+SemRelLM						
UniPRE-large	88.94	82.09	63.79	93.26	89.0	83.42
MultiPRE-large	91.67	90.8	67.82	94.74	91.0	87.21
SemPRE-large	80.13	72.14	46.55	93.26	74.0	73.22

Micro F1-score (in %) on lexical relation classification. RelBERT results are from (Ushio et al., 2023)

Scan QR to check more results and visualizations.



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[1] Asahi Ushio, Jose Camacho-Collados, and Steven Schockaert. 2023. Relbert: Embedding relations with language models. Preprint, arXiv:2310.00299.

